

INTRODUCTION TO INFORMATION SECURITY

Play Fair Decryption and Vernam Cipher



Course Objectives

The key objectives of this course include:

1. The purpose of this course is to understand the information security foundations, security design principles, security mechanisms, symmetric and asymmetric cryptography, encryption, hash functions, and digital signatures.
2. It will introduce authentication and access control; software security; vulnerabilities and protections; malware; database security; network security; firewalls; and intrusion detection.
3. It will introduce security policies, policy formation and enforcement, risk assessment, cybercrime, law and ethics in information security, privacy, and the anonymity of data.

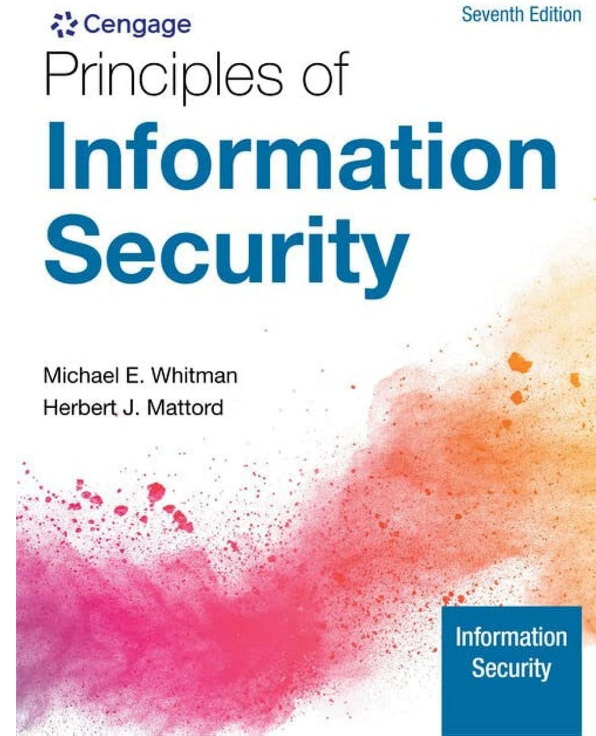
Course Learning Outcomes

After successful completion of this course, the students should be able to:

- | | GA | BT Level |
|--|----|----------|
| 1. Explain key information security concepts and legal, ethical, and professional issues in information security. | 2 | C2 |
| 2. Apply various security and risk management tools for achieving information security and privacy. | 3 | C3 |
| 3. Identify appropriate cryptography techniques to tackle and solve problems in the discipline of information security | 4 | C4 |

Mapping of GA's/PLO's with CLO's

PLOs/CLOs	CLO1	CLO2	CLO3
PLO1 (Academic Education)			
PLO2 (Knowledge for Solving Computing Problems)	✓		
PLO3 (Problem Analysis)		✓	
PLO4 (Design/Development of Solutions)			✓
PLO5 (Modern Tool Usage)			
PLO6 (Individual and Teamwork)			
PLO7 (Communication)			
PLO8 (Computing Professionalism and Society)			
PLO9 (Ethics)			
PLO10 (Life-long Learning)			



Recommended Textbook and Reference Books

Text Books:

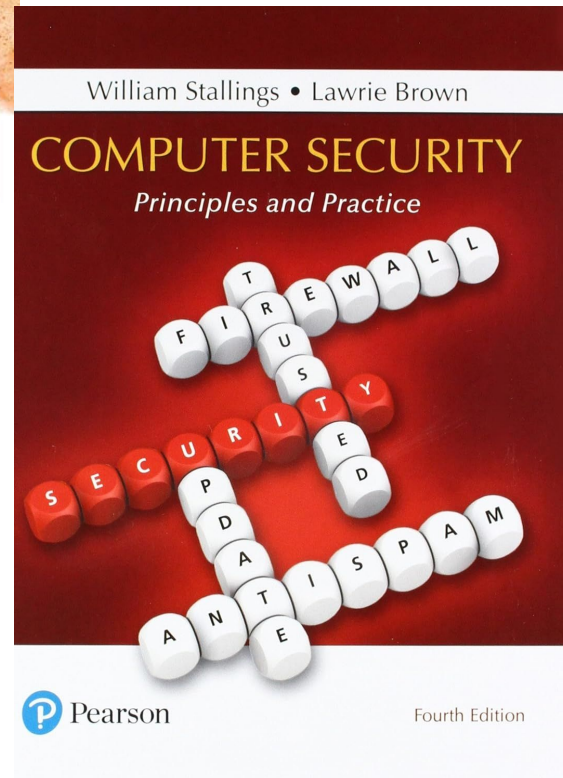
1. Computer Security: Principles and Practice, by William Stallings, 3rd edition 2014

Reference:

1. Principles of Information Security by M. Whitman and H. Mattord, 6th edition, 2017

2. Computer Security by Dieter Gollmann, 3rd edition, 2010

3. Computer Security Fundamentals by William Easttom, 3rd edition, 2016



Playfair Ciphers Decryption

Decrypting the Playfair cipher is as simple as doing the same process in reverse. The receiver has the same key and can create the same key table, and then decrypt any messages made using that key.

Key: monarchy

ciphertext: gatlmzclrqtx

Playfair Cipher Decryption Algorithm:

The Algorithm consists of 2 steps:

1. Generate the key Square(5×5) at the receiver's end:

1. The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.
2. The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.

Playfair Cipher Decryption Algorithm:

2. Algorithm to decrypt the ciphertext: The ciphertext is split into pairs of two letters (digraphs).

Note: The ciphertext always have even number of characters.

For example:

CipherText: "gatlmzclrqtx"

After Split: 'ga' 'tl' 'mz' 'cl' 'rq' 'tx'

Rules for Encryption

If both the letters are in the same column: Take the letter above each one (going back to the bottom if at the top).

For example:

Diagraph: "cl"

Decrypted Text: me

Decryption:

c -> m

l -> e

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rules for Encryption

If both the letters are in the same row: Take the letter to the left of each one (going back to the rightmost if at the leftmost position).

For example:

Diagraph: "tl"

Decrypted Text: st

Decryption:

t -> s

l -> t

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rules for Encryption

If neither of the above rules is true: Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

For example:

Diagraph: "rq"

Decrypted Text: nt

Decryption:

r -> n

q -> t

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Plain Text: "instrumentsz"

Encrypted Text: gatlmzclrqtz

Decryption:

(red)-> (green)

ga -> in

tl -> st

mz -> ru

cl -> me

rq -> nt

tx -> sz

in:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

st:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

ru:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

me:

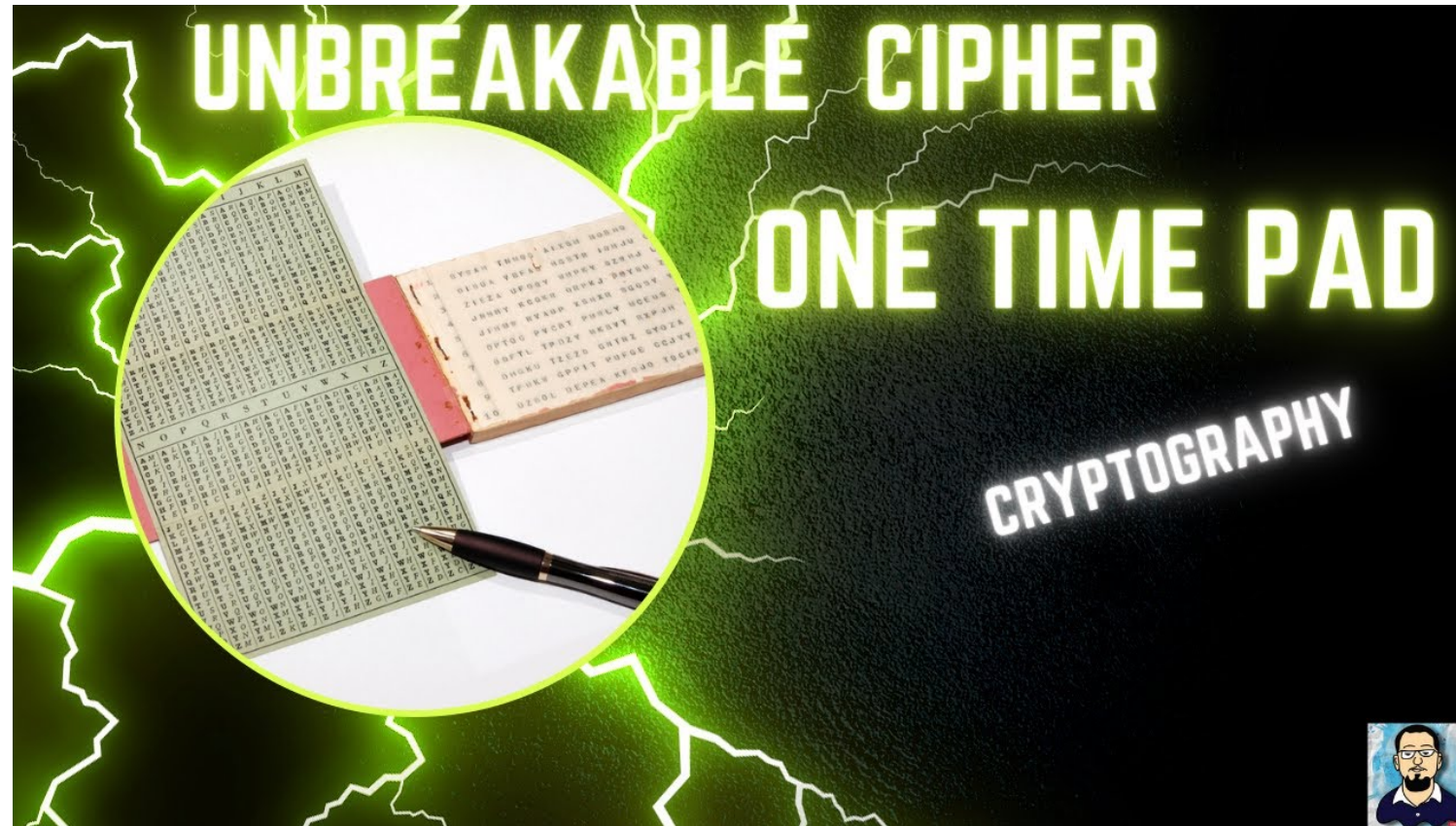
M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

nt:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

sz:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z



VERNAM CIPHER

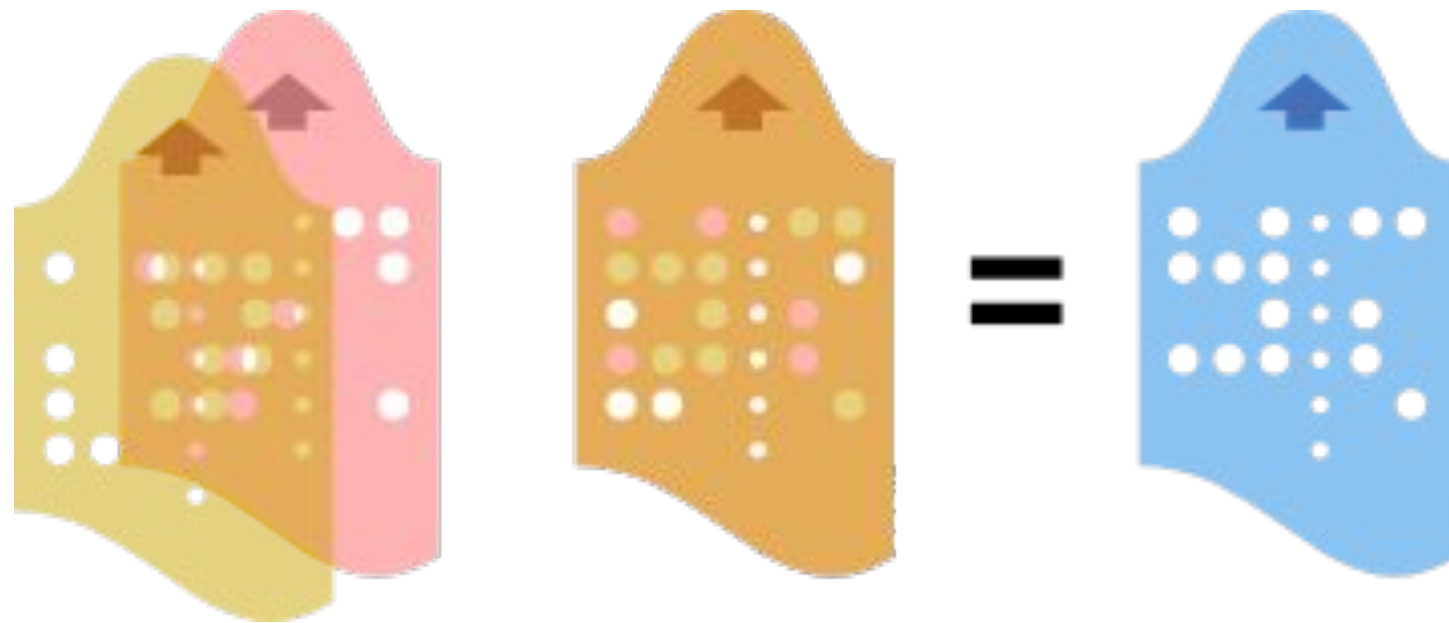
Vernam Cipher

- Vernam Cipher, also known as the one-time-pad.
- Gilbert Vernam invented and patented his cipher in 1917 while working at AT&T.
- Vernam cipher Also known as One-time-pad.

What Is One-Time pad?

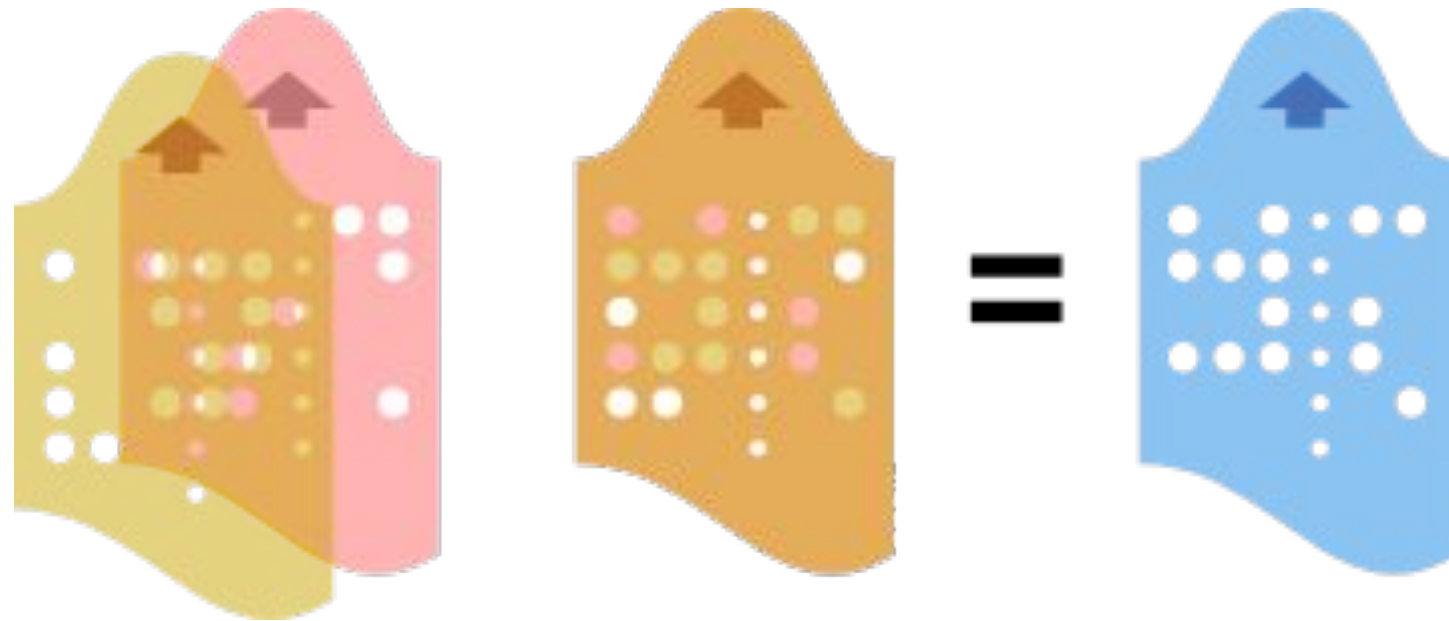
- In cryptography, the one-time pad is an encryption technique that cannot be cracked, but requires the use of a one-time pre-shared key the same size as the message being sent.
- In this technique, a plaintext is paired with a random secret key (also referred to as a **one-time pad**)

Encryption



Plaintext + key = cipher text

Decryption



Cipher text - key = Plain text

Some Rules for Encryption

- First We chose plain text which we want to convert into cipher text.
- We can chose random key.
- Key length is always equal to length of cipher text.
- After adding plain text and keys .If num is ≥ 26 then we subtract 26 from cipher text in Encryption.
- Keys have two copies One for sender and one for receiver.
- Keys is discarded after one time use.

TABLE ALPHABET



A	0	I	8	Q	16	Y	24
B	1	J	9	R	17	Z	25
C	2	K	10	S	18		
D	3	L	11	T	19		
E	4	M	12	U	20		
F	5	N	13	V	21		
G	6	O	14	w	22		
H	7	P	15	X	23		



Encryption Example

Plain-text: H E L L O

Random-Key=G H A U P

Now check the values from table both plain-text and key:

H:7 E:4 L:11 L:11 O:14

G:6 H:7 A:0 U:20 P:15

Now using formula of Encryption:

- plaintext + key = cipher text

H:7 E:4 L:11 L:11 O:14

+

G:6 H:7 A:0 U:20 P:15

13 11 11 31 29

13:N 11:L 11:L 5:F 3:D

N L L F D is cipher text

Decryption

N L L F D is cipher text that is send by sender

Cipher text – key

N:13 L:11 L:11 F:5 D:3

-

G:6 H:7 A:0 U:20 P:15

7 4 11 -15 -12

! Now for negative values we add 26 to make it positive.

7	4	11	11	14
H	E	L	<u>L</u>	O

! So we decrypt the same messege at receiver side.

Practice Question

Another Example “same message” but
now this time key is different.

Key = X M C K L

Encryption

```

      H E L L O
7  (H) 4  (E) 11 (L) 11 (L) 14 (O) message
+ 23  (X) 12  (M) 2  (C) 10  (K) 11  (L) key
= 30 16 13 21 25 message + key
= 4  (E) 16  (Q) 13  (N) 21  (V) 25  (Z)  (message + key)

E Q N V Z → ciphertext
```

Decryption:

E Q N V Z cipher-text

4 (E) 16 (Q) 13 (N) 21 (V) 25 (Z) cipher-text

- 23 (X) 12 (M) 2 (C) 10 (K) 11 (L)

key = -19 4 11 11 14

For negative value we add 26 for make it positive

7 (H) 4 (E) 11 (L) 11 (L) 14 (O) cipher-text - key

H E L L O → message

Encryption Algorithm (XOR)

- ❑ Assign a number to each character of the plain text and the key according to alphabetical order.
- ❑ Convert that number into binary both for the plain text and key
- ❑ Bitwise XOR both the number (Corresponding plain-text character number and Key character number).
- ❑ Subtract the number from 26 if the resulting number is greater than or equal to 26, if it isn't then leave it.

Encryption Algorithm (XOR)

Plain-Text: O A K

Key: S O N

O ==> 14 = 0 1 1 1 0

S ==> 18 = 1 0 0 1 0

Bitwise XOR Result: 1 1 1 0 0 = 28

Encryption Algorithm (XOR)

Since the resulting number is greater than 26, subtract 26 from it. Then convert the Cipher-Text character number to the Cipher-Text character.

28 - 26 = 2 ==> C

CIPHER-TEXT: C

Similarly, do the same for the other corresponding characters

PT: O A K

NO: 14 00 10

KEY: S O N

NO: 18 14 13

Encryption Algorithm (XOR)

New Cipher-Text is after getting the corresponding character from the resulting number.

CT-NO: 02 14 07

CT: C O H

Decryption Algorithm (XOR)

- ❑ Convert the Cipher-Text letters to numbers (A=0, B=1, ..., Z=25):
 - ❑ C $\rightarrow$ 2
 - ❑ O $\rightarrow$ 14
 - ❑ H $\rightarrow$ 7

- ❑ Convert the Key letters to numbers:
 - ❑ S $\rightarrow$ 18
 - ❑ O $\rightarrow$ 14
 - ❑ N $\rightarrow$ 13

Decryption Algorithm (XOR)

- ❑ Perform the reverse XOR operation:
- ❑ $PT = (CT \oplus KEY) \bmod 26$

- ❑ For C (2) and S (18):
- ❑ $PT = (2 \oplus 18) \bmod 26$
 - ❑ Convert to binary:
 - ❑ $2 = 00010$
 - ❑ $18 = 10010$
 - ❑ XOR result: $10000 = 16 (O)$

Decryption Algorithm (XOR)

- ❑ For O (14) and O (14):
 - ❑ $PT = (14 \oplus 14) \bmod 26$
 - ❑ Convert to binary:
 - ❑ $14 = 01110$
 - ❑ $14 = 01110$
 - ❑ XOR result: $00000 = 0$ (A)

Decryption Algorithm (XOR)

- ❑ For H (7) and N (13):
- ❑ $PT = (7 \oplus 13) \bmod 26$
- ❑ Convert to binary:
 - ❑ $7 = 00111$
 - ❑ $13 = 01101$
 - ❑ XOR result: $01010 = 10$ (K)
- ❑ Convert the numbers back to letters: $16 \rightarrow O, 0 \rightarrow A, 10 \rightarrow K$
- ❑ Thus, the Decrypted Plain-Text is O A K, which matches the original message.